

FINITENESS OF PSEUDO-ANOSOV FLOWS WITH NO PERFECT FITS (PART 2)

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Theorem 1 (Barthelmé-T.-Zung). *Every closed 3-manifold M admits at most finitely many pseudo-Anosov flows with no perfect fits up to orbit equivalence.*

Proof strategy: Fix a closed oriented 3-manifold M . Let ϕ be a pseudo-Anosov flow with no perfect fits on M .

- (1) Li, in preparation: There are at most finitely many possible isotopy classes of $\text{sing}(\phi)$.
- (2) T. 2024: ϕ admits a Birkhoff section whose negative boundary components lie along $\text{sing}(\phi)$. In other words, $\phi^\circ = \phi|_{M \setminus \text{sing}(\phi)}$ admits a positive Birkhoff section.
- (3) Show that ϕ° can be blown up into a Reeb flow of a tight contact structure ξ with no Giroux torsion in $M \setminus \nu(\text{sing}(\phi))$, c.f. Zung 2025.
- (4) Use cylindrical contact homology to show that ξ determines the **spectrum** $\mathcal{P}(\phi)$, i.e. the set of free homotopy classes of closed orbits of ϕ , c.f. Barthelmé-Bowden-Mann 2024.
- (5) Colin-Giroux-Honda 2009: Every 3-manifold has at most finitely many tight contact structures with no Giroux torsion, up to contactomorphism.
- (6) Barthelmé-Frankel-Mann 2025: $\mathcal{P}(\phi)$ determines ϕ up to isotopic equivalence.

Step 1: Controlling $\text{sing}(\phi)$.

Theorem 2 (Gabai 1999). *Let M be a closed orientable atoroidal 3-manifold. There exists a finite collection of essential branched surfaces $B_1, \dots, B_n$ such that every essential lamination is carried by some B_k .*

Theorem 3 (Li, in preparation). *Let M be a closed orientable atoroidal 3-manifold. There are at most finitely many possibilities for the isotopy class of $\text{sing}(\phi)$ and the degeneracy curves for a pseudo-Anosov flow ϕ on M .*

Step 2: Constructing a Birkhoff section that is positive away from $\text{sing}(\phi)$.

A **Birkhoff section** to a flow ϕ is a cooriented surface with boundary S where

- the interior of S is positively transverse to ϕ ,
- the boundary of S is tangent to ϕ , and
- every orbit intersects S in finite forward and backward time.

The last property allows us to talk about the first return map $f : \text{int}(S) \rightarrow \text{int}(S)$. When ϕ is pseudo-Anosov, f is a pseudo-Anosov map. The restriction of ϕ to $M \setminus \partial S$ is orbit equivalent to the suspension flow ϕ_f on the mapping torus M_f . Conversely, ϕ can be recovered from ϕ_f by Dehn filling the cusps.

Suppose M is oriented, then each boundary component c of S inherits one orientation as a boundary component of S and one orientation as a closed orbit of ϕ . We say that c is **positive** if the two orientations agree, otherwise we say that c is **negative**.



Theorem 4 (T. 2024). *Let ϕ be a pseudo-Anosov flow with no perfect fits on a closed oriented 3-manifold M . Then ϕ admits a Birkhoff section with boundary components along $\text{sing}(\phi)$ and one other closed orbit.*

Step 3: Blowing up ϕ into a Reeb flow.

The key idea is to pass through a **stable Hamiltonian structure**. This is a pair consisting of a 2-form ω and a 1-form λ satisfying the following properties:

- $d\omega = 0$,
- $\ker(\omega) \subset \ker(d\lambda)$, i.e. $d\lambda = F\omega$ for some function F , and
- $\lambda \wedge \omega > 0$.

Examples:

- (1) Suppose $f : S \rightarrow S$ is a surface diffeomorphism that preserves an area form Ω . Let λ be the 1-form dt on the mapping torus M_f . Let ω be the pull-back of Ω over projection along t . Then (ω, λ) is a SHS. Here $F = 0$.
- (2) Suppose λ is a positive contact form. Then $(d\lambda, \lambda)$ is a SHS. Here $F = 1$.

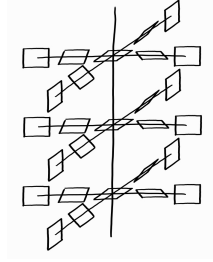
Our SHS will be a mix of these two examples.

Let S be a Birkhoff section as in [Theorem 4](#). Let $f : \text{int}(S) \rightarrow \text{int}(S)$ be the first return map. Since all singular orbits lie in ∂S , f is smooth and preserves an area form Ω . Define a SHS on $M \setminus \partial S \cong M_f$ as in (1) above.

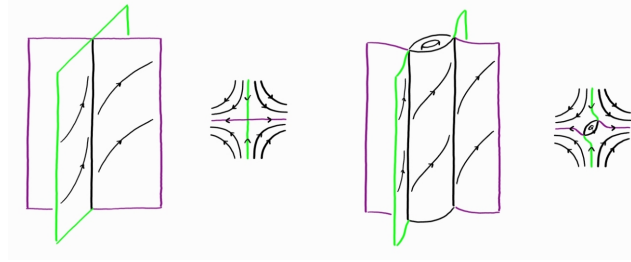
Extend (ω, λ) over the positive nonsingular boundary orbits of S . Here the standard model is

$$\begin{aligned}\lambda &= r^2 d\theta + dz \\ \omega &= r dr d\theta\end{aligned}$$

in cylindrical coordinates (r, θ, z) on the solid torus. Then (ω, λ) is a SHS that is foliation-like away from the positive boundary orbits and contact-like near the positive boundary orbits.



The next step is to diffuse the contact-like region. First notice that if we blow up a closed orbit γ of ϕ into invariant torus shells, then the blown-up first return map still preserves an area form Ω , but now there is an infinite-dimensional space of choices for Ω given by adding mass to the invariant circles. In particular, one can vary the (co)homology class of ω by adding positive multiples of $[\gamma]$.



Proposition 5. *Let ϕ be a pseudo-Anosov flow with no perfect fits on a closed 3-manifold M . Then $\{[\gamma] \in H_1(M^\circ, \partial M^\circ) \mid \gamma \text{ is a closed orbit of } \phi\}$ generates $H_1(M^\circ, \partial M^\circ)$ as a cone.*

Proof. Suppose otherwise, then there exists $\alpha \in H^1(M^\circ, \partial M^\circ) \cong H_2(M^\circ)$ that pairs nonnegatively with every closed orbit γ . A theorem of Mosher implies that α can be realized as a surface S in M° almost transverse to ϕ . Since S is disjoint from $\text{sing}(\phi)$, S must be a torus. But pseudo-Anosov flows with no perfect fits do not have almost transverse tori, contradiction. $\square$

Thus we can blow up at enough orbits to make ω exact, say $\omega = d\beta$.

Now consider the 1-form $\alpha = \lambda + \varepsilon\beta$. We have $d\alpha = d\lambda + \varepsilon d\beta = (F + \varepsilon)\omega$. Thus $\alpha \wedge d\alpha = (F + \varepsilon)(\lambda + \varepsilon\beta) \wedge \omega$. Since $\lambda \wedge \omega > 0$, we have $\alpha \wedge d\alpha > 0$ for small $\varepsilon > 0$. Moreover, since $d\alpha$ is proportional to ω , ϕ is (a reparametrization of) the Reeb flow of α .

Since ϕ has no nullhomotopic closed orbits, $\xi = \ker \alpha$ is hypertight, thus tight.

If ξ has Giroux torsion, then the torsion must occur along a boundary torus T . By a result of Chaidez-Pan, if this is the case, then essentially every homotopy class on T is realized by some closed orbit of ϕ , which is not the case.

We also further cap off ξ by a standard piece at each boundary torus, so that we have the adapted boundary condition for contact homology, and so that we can apply the precise statement of Colin-Giroux-Honda 2009.

Step 4: Recovering free homotopy data from the contact structure.

Let α be a contact form for a contact structure ξ . Suppose that the Reeb flow R of α is nondegenerate and has no nullhomotopic closed orbit. Here, a closed orbit γ is **nondegenerate** if the differential of its first return map does not have 1 as an eigenvalue. In that case, γ is either elliptic, positive hyperbolic, or negative hyperbolic. A flow is **nondegenerate** if all of its closed orbits are nondegenerate. It is a fact that the Reeb flow of a generic contact form to a fixed contact structure is nondegenerate.

The cylindrical contact chain complex $CC(\alpha)$ is the $\mathbb{F}_2$ vector space generated by the closed orbits of R , with differential defined by counting holomorphic cylinders in $\mathbb{R} \times M$. The differential ∂ preserves the grading of $CC(\alpha)$ by free homotopy class, and lowers the $\mathbb{Z}/2$ grading by Conley-Zehnder index by 1. Here, the **Conley-Zehnder index** of γ is

$$\text{ind}_{\text{CZ}}(\gamma) = \begin{cases} 0 & \text{if } \gamma \text{ is positive hyperbolic,} \\ 1 & \text{if } \gamma \text{ is elliptic or negative hyperbolic.} \end{cases}$$

In particular, the Lefschetz index of γ is $\text{ind}_{\text{Lef}}(\gamma) = -(-1)^{\text{ind}_{\text{CZ}}(\gamma)}$.

Theorem 6 (Hutchings-Nelson 2016). *The homology of $CC(\alpha)$ is determined as a vector space bigraded by free homotopy class and Conley-Zehnder index by the contact structure ξ only.*

We write $HC(\xi)$ for this homology. In particular, in a fixed grading of free homotopy class g , the Euler characteristic over Conley-Zehnder index $\chi(HC(\xi; g)) = \chi(CC(\xi; g)) = -\sum_{[\gamma]=g} \text{ind}_{\text{Lef}}(\gamma)$. The latter is unchanged by local perturbations of the flow.

In the setting of **Theorem 1**, each nonperipheral free homotopy class of M° contains at most one closed orbit of ϕ° , thus

$$\sum_{[\gamma]=g} \text{ind}_{\text{Lef}}(\gamma) = \begin{cases} 1 & \text{if } g \text{ contains an orientation-preserving closed orbit,} \\ -1 & \text{if } g \text{ contains an orientation-reversing closed orbit,} \\ 0 & \text{if } g \text{ does not contain a closed orbit.} \end{cases}$$

In particular, the nonperipheral spectrum of ϕ° is simply the support of $HC(\xi)$, thus is completely determined by ξ .

The spectrum of ϕ is just the union of the image of the nonperipheral spectrum of ϕ° , and $\text{sing}(\phi)$, so it is determined by ξ as well.

Step 5: Finiteness of contact structures.

Theorem 7 (Colin-Giroux-Honda 2009). *Let Q be a compact 3-manifold, possibly with boundary and ξ the germ of a contact structure along ∂Q . Then up to the action of diffeomorphisms isotopic to the identity rel boundary and Dehn twists along tori in the interior of Q , for any fixed n , there are a finite number of tight contact structures which are equal to ξ near ∂Q and which do not contain $2\pi n$ Giroux torsion.*

Step 6: Recovering the flow from the spectrum.

Theorem 8 (Barthelmé-Frankel-Mann 2025). *Let ϕ_1 and ϕ_2 be two pseudo-Anosov flows on a closed 3-manifold M . Suppose ϕ_1 has no perfect fits and $\mathcal{P}(\phi_1) = \mathcal{P}(\phi_2)$, then ϕ_1 and ϕ_2 are isotopic equivalent.*

More generally, if the assumption of no perfect fits is removed, then $\mathcal{P}(\phi)$ determines ϕ up to finitely many possibilities.

Future directions.

- **Orbit space characterization of +BAS.**

The proof actually works more generally to show finiteness of pseudo-Anosov flows on atoroidal manifolds that admit a Birkhoff section that is positive away from singularities (+BAS).

Question 9. Is there an orbit space characterization of +BAS?

- **Other rigid orbits.**

Theorem 3 says that the singular orbits of a pseudo-Anosov flow are rigid.

Conjecture 10. *Branching orbits are also rigid.*

- **Direct proof from triangulations.**

The proof of both **Theorem 2** and **Theorem 7** involve putting objects in ‘normal form’ with respect to a fixed triangulation of M .

Question 11. Is there a proof of **Theorem 1** that just involves putting pseudo-Anosov flows in ‘normal form’ with respect to a fixed triangulation?

Due to their method of proof, it should be possible to make **Theorem 2** and **Theorem 7** effective as an exponential bound in the number of tetrahedra.

Question 12. Can one give an explicit exponential bound in the number of tetrahedra for **Theorem 1**?